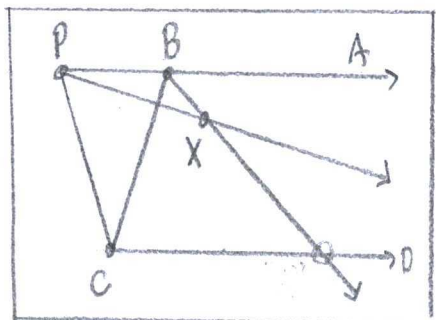


Chapter 6: The Discovery of Non-Euclidean Geometry

Major Exercises:

1.

(from exercise) "If $P * B * A$ or
if $B * P * A$, then
 $\overline{PA} \parallel \overline{CD}$."



"Biangle"

$\circ$ = Ray $\overrightarrow{BA}$ limiting
parallel to $\overrightarrow{CD}$

Proof Hypothesis

Evidence.

RAA hypothesis

1) From data provided
in figure.

Case 1: $P * B * A$

A) $\overrightarrow{PX}$ intersects $\overrightarrow{CD}$
because $\overrightarrow{PX}$ is
an interior ray
and not parallel
to CD .

B) $\overline{PA} \parallel \overline{CD}$

(Pg 276) "Definition-
Interior Ray"

(Pg 20) "Definition-
Parallel"

(Pg 276) "Definition-
Biangle"

Case 2: $B * P * A$

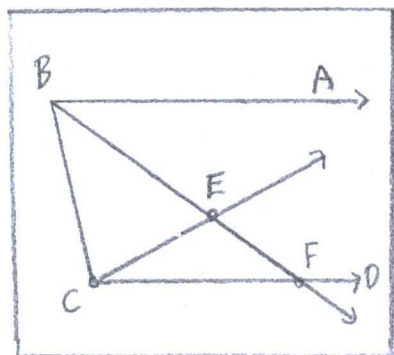
A) $\overrightarrow{PX}$ intersects $\overrightarrow{CD}$
because $\overrightarrow{PX}$ is
an interior ray
and not parallel
to CD .

B) $\overrightarrow{PA} \parallel \overrightarrow{CD}$

(pg 276) "Definition-
Interior Ray"

(pg 276) "Definition-
Biangle"

2.



"Biangle"

(from exercise) "If $\overrightarrow{BA} \parallel \overrightarrow{CD}$, then $\overrightarrow{CD} \parallel \overrightarrow{BA}$."

Proof: Hypothesis

Evidence

1) An assumption,

False RAA hypothesis #1

[ABCD is not
closed at C

2) The interior ray

False RAA hypothesis #

$\overrightarrow{CE}$ does not
intersect $\overrightarrow{BA}$

3) Point E, which is

Corollary to

just a label, can
be chosen so that

$$\angle BEC < \angle ECD$$

4) Segment $\overline{BE}$ does
not intersect $\overline{CD}$

5) Interior Ray $\overrightarrow{BE}$
intersects CD in
a point F , and
 $B * E * F$.

6) Since $\angle BEC$ is
an exterior angle
for $\triangle EFC$, we
have $\angle BEC > \angle ECF$

7) Contradiction

Aristotle's Axiom

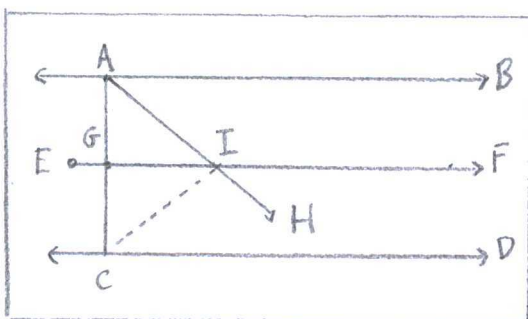
(pg. 276) "Definition-
Interior"

(pg. 276) "Definition-
B, angle"

Exterior Angle Theorem
4.2

False $\rightarrow$ False $\rightarrow$ True
is True!

3.



"limiting Parallel"

(from exercise) Transitivity of
Limiting Parallel

"If $\overline{AB}$ and $\overline{CD}$ are both
limiting parallel to $\overline{EF}$, then

are limiting parallel to each other"

Proof: Hypothesis

Evidence

1) $\vec{AB}$ and $\vec{CD}$ have
no point in common

Euclid's Postulate V

2) Hence two cases
depending on whether
 $\vec{EF}$ between $\vec{AB}$
and $\vec{CD}$ or $\vec{AB}$ and
 $\vec{CD}$ on the same
side of $\vec{EF}$.

RAA hypothesis

3) In the case
where $\vec{EF}$ between
 $\vec{AB}$ and $\vec{CD}$, G
is the intersection
of $\vec{AC}$ with $\vec{EF}$

Proposition 2.1

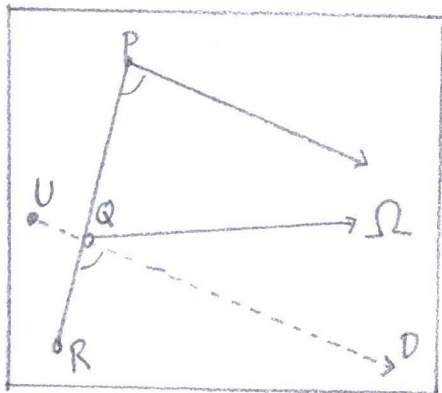
4) Any ray $\vec{AH}$
interior to $\angle GAB$
must intersect

(pg 276) "Definition-
Interior ray"

- EF in a point I.
- 10) Since CD enters $\triangle AEI$ at G and does not intersect side EI, it must intersect AI
- 11) Therefore, CD limits parallel to AB

Pasch's Theorem

RAA result



(From exercise) "If $\triangle PQR$ is a singly asymptotic triangle, the exterior angles at P and Q are greater than their respective opposite interior angles."

Proof: Hypothesis

Evidence

1) Given $R \times Q \times P$. We must show that $\angle RQ\Omega$ is greater than $\angle QP\Omega$

RAA hypothesis

2) Let $\vec{QD}$ be a unique ray on the same side

Axiom B-4

of $\overleftrightarrow{PQ}$ as ray $\overrightarrow{Q\Omega}$
 such that $\angle RQD \cong$
 $\angle QP\Omega$

3) If $U \times Q \times D$, then
 $\angle UQP \cong \angle QP\Omega$

4) $\overleftrightarrow{QD}$ is divergently
 parallel to $\overleftrightarrow{P\Omega}$

5) Hence $\overleftrightarrow{QD}$ is
 between $\overleftrightarrow{QR}$ and
 $\overleftrightarrow{Q\Omega}$

6) $\angle RQ\Omega > \angle QP\Omega$

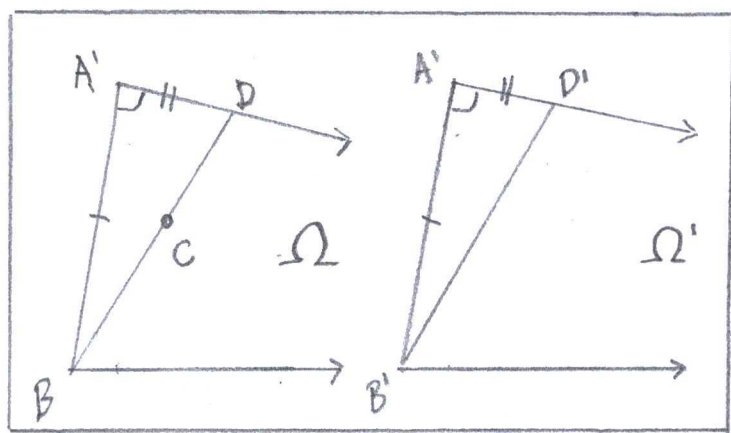
Axiom C-4

(pg 199) "Definition
 Corresponding
 angles"

(pg 115) "Definition-
 between"

RAA statement

5.



"Congruence Theorem"

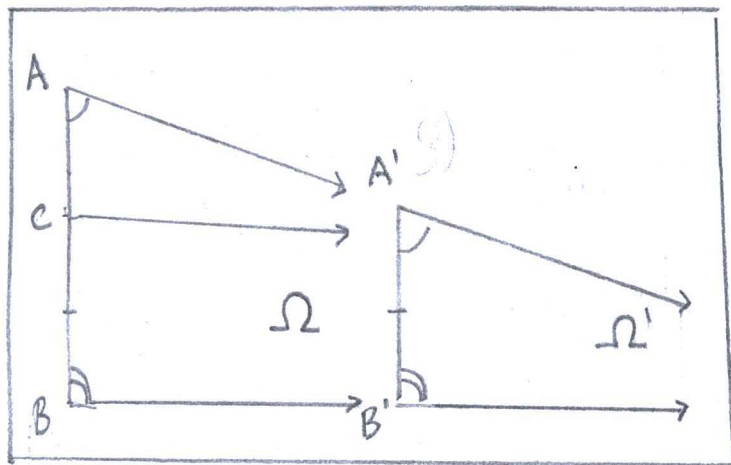
(from exercise) 'If a asymptotic triangles $\Delta AB\Omega$ and $\Delta A'B'\Omega'$, we have $\angle B A \Omega \cong \angle B' A' \Omega'$, then $\angle A B \Omega \cong \angle A' B' \Omega'$, if and only if $\overline{AB} \cong \overline{A'B'}$ '

Proof: Hypothesis	Evidence
$\overline{AB} \cong \overline{A'B'}$ and $\angle A B \Omega > \angle A' B' \Omega'$	RAA conjecture #1 contrary
2) Unique ray $\overrightarrow{BC}$ is between $\overrightarrow{B\Omega}$ and $\overrightarrow{BA}$, such $\angle ABC \cong \angle A'B'\Omega'$	(Pg 115) "Definition- between"
3) $\overrightarrow{BC}$ intersects $\overrightarrow{A\Omega}$ at a point D.	Proposition 3.9(b)
4) D' be a unique point on $A'\Omega'$, such $\overline{AD} \cong \overline{A'D'}$	Axiom C-1
5) $\Delta BAD \cong \Delta B'A'D'$	Axiom C-6 (S.A.S.)

$$6) \angle ABD \cong \angle A'B'\Omega'$$

$$7) \angle AB\Omega \cong \angle A'B'\Omega'$$

and $\overline{A'B'} < \overline{AB}$



8) If C is a point on $\overline{AB}$ such that $\overline{BC} \cong \overline{B'A'}$, then the ray $\overrightarrow{C\Omega}$ is the limiting parallel to $\overrightarrow{A\Omega}$

9) Then $C\Omega$ is also the limiting parallel to $\overrightarrow{B\Omega}$

10) In the case, $\angle BC\Omega \cong \angle B'A'\Omega'$

11) $\angle BC\Omega > \angle BA\Omega$

in false result.

So, $\angle BC\Omega \cong \angle BA\Omega$ and

Absurd! RAA

(pg 260) "Definition-
Angles of
Parallelism"

False RAA conjecture

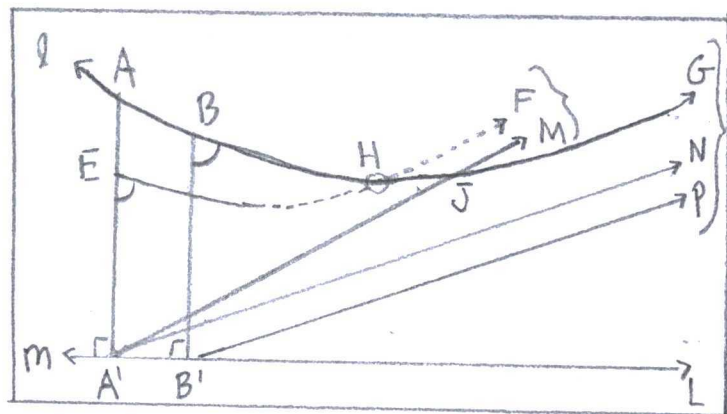
(pg 276) "Definition-
Limiting Parallel"

Axiom C-4

False RAA result

triangles $\triangle ABQ$ and $\triangle A'B'Q'$
limit parallels by
asymptotes.

6.



(From exercise) Conclusion of
the proof of
Theorem 6.3

" $\overrightarrow{EF}$ intersects $\overrightarrow{AG}$ "

"Theorem 6.3" Proof Hypothesis

Evidence

1) $\overrightarrow{A'M}$ limits parallel
to $\overrightarrow{EF}$, $\overrightarrow{A'N}$ limits
parallel to $\overrightarrow{AG}$, and
 $\overrightarrow{B'P}$ limits parallel
to $\overrightarrow{BG}$.

RAA hypothesis

2) Since $\overrightarrow{EA'} \cong \overrightarrow{BB'}$ and
 $\angle A'EF \cong \angle B'BG$, we
have $\angle EA'M \cong \angle BB'P$

Theorem 6.3
because congruent
Saccheri quadrilaterals

3) $\overrightarrow{B'L}$ differs from $\overrightarrow{B'P}$
and $\overrightarrow{A'L}$ differs
from $\overrightarrow{A'N}$.

(pg 18) "Definition-
Emanating from
vertex"

4) $\angle MA'L \cong \angle PB'L$

5) $\overrightarrow{BP}$ limits parallel
to $\overrightarrow{A'N}$

6) Hence, $\angle NA'L$ is
smaller than $\angle PB'L$

7) $\overrightarrow{A'M}$ lies between
 $\overrightarrow{A'N}$ and $\overrightarrow{A'A}$, and
intersects $\overrightarrow{AG}$ in a
point J

8) J is on the same
side of $\overleftrightarrow{EF}$ as A'
hence the side opp-
osite from A.

9) Thus, $\overrightarrow{AJ}$ intersects
 $\overleftrightarrow{EF}$ in a point H,
on $\overleftrightarrow{EF}$ because
same side to $\overrightarrow{AA'}$
as J.

(pg 171) "Definition-
Complementary
angles"

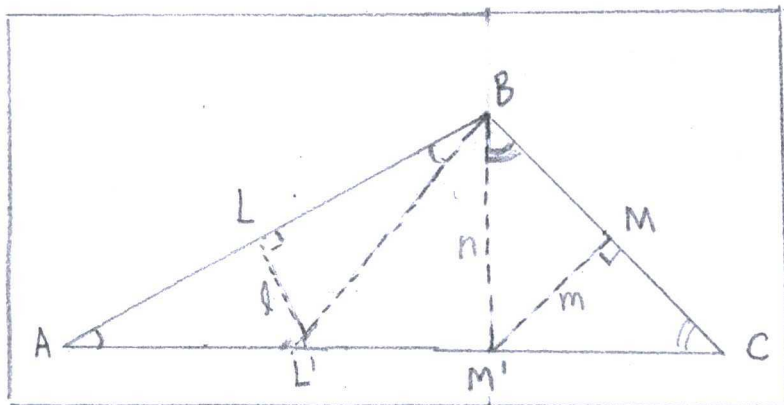
(pg 276) "Definition-
Limiting Parallel"

Contrary to
Proposition 6.6

Proposition 3.9(b)

Axiom B-4(ii)

RAA result



(pg 281) Lemma 6.1

"Given $\triangle ABC$. Let l, m , and n be the perpendicular bisectors of sides $\overline{AB}, \overline{BC}$, and $\overline{AC}$ at their midpoints L, M , and

N , respectively. Let $\overline{AC} \geq \overline{AB}$ and $\overline{AC} \geq \overline{BC}$ (AC is the longest side). Then, l, m , and n intersect AC ."

(from exercise) "If perpendicular bisectors l and m are asymptotically parallel in the direction of an ideal point Ω , then the third perpendicular bisector n is asymptotically parallel to l and m in the same direction Ω "

Proof Hypothesis	Evidence
1) $(\angle B)^\circ \geq (\angle A)^\circ$ and $(\angle B)^\circ \geq (\angle C)^\circ$	RAA conjecture
2) Hence point L' is on $\overline{AC}$ such $\angle A \cong \angle L'BA$, and a point M' on $\overline{AC}$ such	Axiom C-4

$$\angle C \cong \angle M'BC$$

3) Then $\overline{AL'} \cong \overline{BL'}$ and
 $\overline{CM'} \cong \overline{BM}$

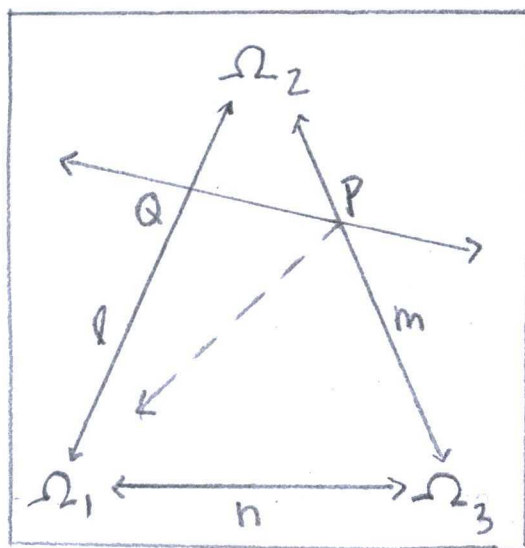
4) Thus l is the line
 from L to L' , and
 $m = \overline{MM'}$

5) All three perpendicular
 bisectors cut
 $\overline{AC}$.

Converse to
 Proposition 3.10

Proposition 4.4(b)
 (Bisectors)

RAA result.



(pg 292) Lemma 6.2

"No line intersects all three
 sides of a trebly asymp-
 totic triangle."

Proof Hypothesis

Evidence

1) Line $\overleftrightarrow{t}$ cuts $\overleftrightarrow{l}$ at Q
 and $\overleftrightarrow{m}$ at P

Proposition 2.1

2) Then ray $\overrightarrow{PQ}$ of $\overleftrightarrow{t}$ lies
 between the rays $\overrightarrow{P\Omega_1}$
 and $\overrightarrow{P\Omega_2}$ which limit

Proposition 3.2

Parallel to l .

3) $\overrightarrow{P\Omega_3}$, the other ray through P limits parallel to n opposite to $\overrightarrow{P\Omega_2}$

4) Hence $\overrightarrow{P\Omega_1}$ lies between $\overrightarrow{PQ}$ and $\overrightarrow{P\Omega_3}$

5) Thus $\overrightarrow{PQ}$ never intersects n

6) Similarly, $\overrightarrow{QP}$ does not intersect n

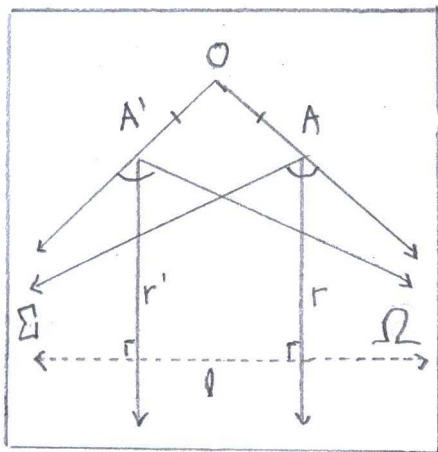
Proposition 3.8(b)

(pg 115) "Definition-between"

Pasch's Theorem

Corollary to Step #5
KAA result

8.



"line of enclosure"

(from exercise) "Given any angle $\angle A'OA$. It is a theorem in hyperbolic geometry that there is a unique line $\overleftrightarrow{l}$ called the line of enclosure of this angle, such that $\overleftrightarrow{l}$ is limiting parallel to both sides $\overline{OA'}$ and $\overline{OA}$ "

Proof Hypothesis

Evidence

$$1) \overrightarrow{OA} \cong \overrightarrow{OA'}$$

2) $\overrightarrow{A\Omega}$ limits parallel ray to $\overrightarrow{OA}$ through A and $\overrightarrow{A\Sigma'}$ limits parallel ray to $\overrightarrow{OA'}$ through A'

3) Rays r and r' are bisectors of $\angle \Sigma A \Omega$ and $\angle \Omega A' \Sigma'$, respect!

4) Rays r and r' have a common perpendicular $\overleftrightarrow{\ell}$

5) $\overrightarrow{OA'}$ is the left limiting parallel, and OA is the right limiting parallel

6) $\angle AOA'$ has a line of enclosure because adjacent rays never intersect, nor asymptotically parallel $\overleftrightarrow{\ell}$

Axiom C-1

(pg 270) "Definition- limiting parallel"

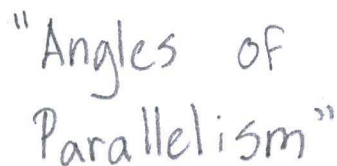
Proposition 4.4(a)
(Bisectors)

Theorem 6.3

(pg 261) "Definition-
Left limiting parallel
&
Right limiting parallel"

Conclusion

(from exercise) "Every acute angle is
an angle of parallelism i.e.



given an acute angle $\angle BOA$,
there is a unique line $\vec{l}$
perpendicular to $\vec{BO}$ and limiting
parallel to $\vec{OA}$

Parallelism	Proof Hypothesis	Evidence
Requirement #1 Theorem 6.1	1) Rectangles do not exist	RAA hypothesis
	2) $\angle BOA$ is acute	(pg 129) "Definition-acute"
	3) B is the foot of the perpendicular from A.	(pg 269) "Definition-medial line"
	4) $\overline{OB} \cong \overline{BB'}$	Euclid's Postulate II
	5) Perpendicular B' intersects $\overline{OA}$ at A	Proposition 2.1
	6) $\angle OA'B' = \angle OAB + \angle AA'B$ $= 2\angle OAB + \angle AA'B$ $> 2\angle OAB$	Proposition 6.1 (Additivity of the defect)

Requirement #2
Theorem 6.2

7) Defect δ increases
by addition per-
pendicular feet from
 $\overline{OB}$ to $\overline{OA}$.

1) A unique point R
on $\overline{OB}$ has vertex
 O opposite to the
rays complement

2) A perpendicular ray
 r emanates from
point R

3) R locates a
ray r

Requirement #3
Theorem 6.3

1) Point C is the
foot to the perpendicular
to $\overline{OA}$ at point D

2) $\overrightarrow{RS}$ is an interior
ray

3) $\overrightarrow{RS}$ intersects OD
in $\triangle OCD$

RAA result

Dedekinds Axiom

(pg 18) "Definition-
Emanates from
Vertex"

(pg 10) "Definition-
emanates from vertex"

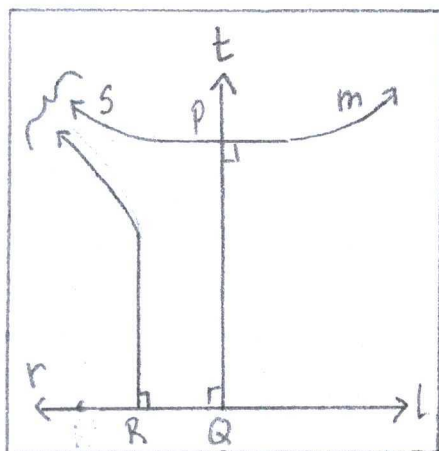
(pg 276) "Definition-
medial line"

(pg 276) "Definition-
interior ray"

Pasch's Theorem

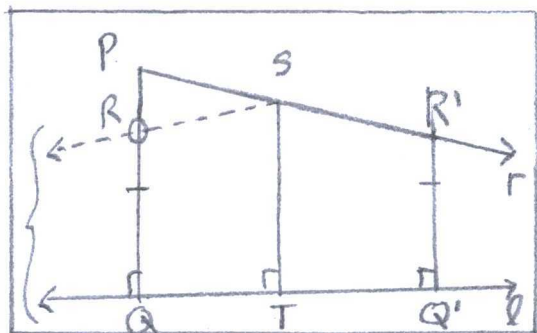
10.

3. (from exercise) "A unique point R on

"Divergently
Parallel"r is perpendicular to l through
R and limits parallel to s "

Proof	Hypothesis	Evidence
1) l and m are		RAA hypothesis
divergently parallel		
2) Line t is a		Proposition 4.1
common perpendicular		
through l at Q		
and m at P .		
3) Ray r emanates		(pg 18) "Definition-
from Q on l		emanating from
		Vertex"
4) The ray s on		(pg 18) "Definition-
m emanates from		emanating from
P .		Vertex"
5) s is on the same		(pg) "Definition-
side t as r		same side"
6) Ray R limits m		RAA result
and also s .		

11



"Asymptotically
Parallel"

(from exercise) "For any point R between P and Q there exists a point R' on ray r such $\overline{R'Q'} \cong \overline{RQ}$, where Q' is the foot of the perpendicular from R' to $\vec{l}$."

Proof: Hypothesis	Evidence
1) Ray r emanates from point P	(pg 18) "Definition-emanates from vertex"
2) r limits parallels to line $\vec{l}$	(pg 276) "Definition-limiting parallel"
3) Point Q is the foot from P to $\vec{l}$	
4) A point R is between Q and P	(pg 115) "Definition-between"
5) A ray from R intersects $\vec{l}$ at point S	Proposition 2.1
6) Point T is the foot of the perpendicular from S to $\vec{l}$	(pg 276) "Definition-medial line"

7) R' is a reflection
across $\overline{ST}$ from R

8) Q' is a reflection
across $\overline{ST}$ from Q

9) The ray opposite
 $\overrightarrow{RS}$ is asymptotically
parallel through a
reflection.

So $\overline{RQ'} \cong \overline{RQ}$

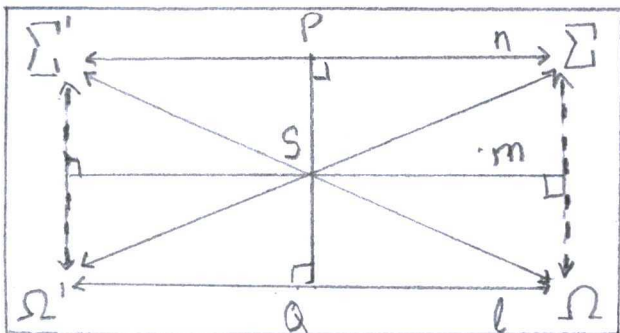
(pg 154) "Definition-
Reflection"

(pg 154) "Definition-
Reflection"

(pg 110) "Definition-
opposite"

12.

a)



(from exercise)

" $\overleftrightarrow{\Omega\Sigma'}$ and $\overleftrightarrow{\Sigma\Omega'}$ meet
at S "

"The symmetry of a
point"

Proof: Hypothesis

Evidence

1) $\overleftrightarrow{l}$ and $\overleftrightarrow{n}$ are di-
vergent parallel
lines

2) $\overline{PQ}$ is a common
perpendicular segment

3) The midpoint S of $\overline{PQ}$

(pg 264) "Definition-
Divergently Parallel"

Proposition 4.1

Proposition 4.3 (midpoint)

is the symmetry
point of l and n .

4) m is a perpendicular
to $\overline{PQ}$ through S

5) Ω and Ω' are
ideal points of l

6) Σ and Σ' are
ideal points of n .

7) P is a midpoint on
 n and Q another
midpoint, but on l

8) $\triangle \Sigma'PS \cong \triangle P\Sigma S \cong \triangle \Omega'QS$
 $\cong \triangle Q\Omega S$

9) $\overleftrightarrow{\Omega\Sigma'} = \overleftrightarrow{\Sigma\Omega'}$ by
congruent triangles

10) $\overleftrightarrow{\Omega\Sigma}$ and $\overleftrightarrow{\Sigma'\Omega'}$
intersect at a point S

b) (from exercise)

" m is perpendicular to both $\overleftrightarrow{\Omega\Sigma}$ and $\overleftrightarrow{\Omega\Sigma'}$ "

(pg 42) "Definition-
Perpendicular"

(pg 279) "Definition-
Ideal Points"

(pg 279) "Definition-
Ideal Points"

Proposition 4.3 (midpoints)

Axiom C-6 (SAS)

(pg 119) "Definition-
Congruent"

Proposition 2.1

Proof Hypothesis

Evidence

1) From the information
shown above

RAA hypothesis

2) $\triangle P\Sigma S$ is a right
triangle, along with
 $\triangle \Sigma'PS$

(pg 19) "Definition-
right triangle"

3) $\triangle \Omega'\Sigma'\Sigma$ is similar
to $\triangle P\Sigma S$

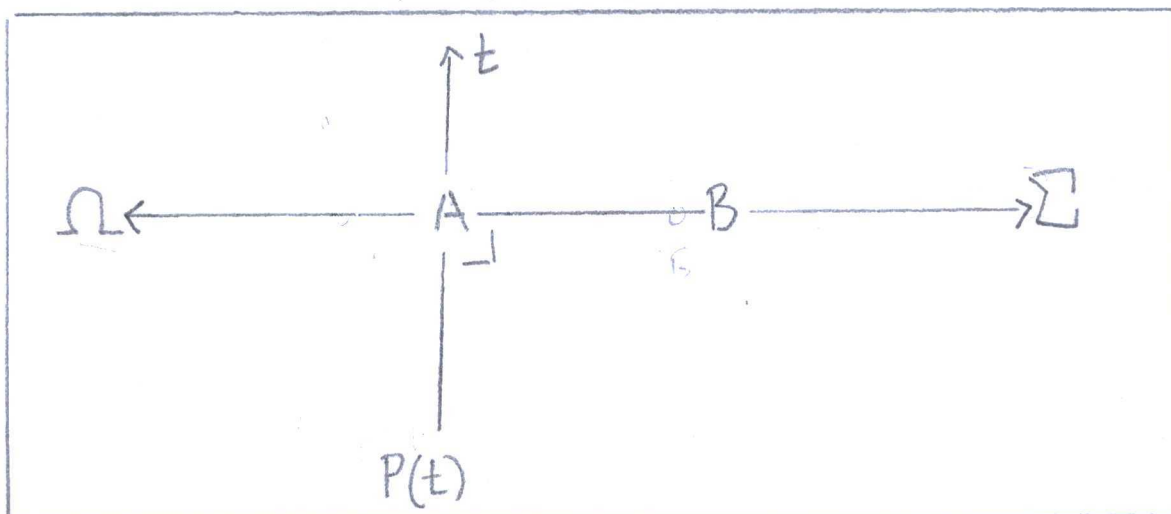
(pg 215) "Definition-
similar"

4) $\triangle \Sigma'PS$ is similar
to $\triangle \Sigma'\Sigma\Omega$

(pg 215) "Definition-
similar"

5) $m \perp \overrightarrow{PQ}$ and $m \perp \overrightarrow{\Omega\Sigma}$
and $m \perp \overrightarrow{\Omega\Sigma'}$

RAA result



$P(t)$ relates
to $\overleftrightarrow{AB}$ and
 $\overleftrightarrow{AB}$ relates
to t , e.g.
isomorphism.

(from exercise) Theorem

$\mathcal{D}$ is a projective plane and P is a polarity
(an isomorphism of $\mathcal{D}$ onto its dual plane).